

Q1: Do #1, #5, #8, #10, 12, 13 in Ch2.

① Find canonical SOP (sum of products) form of the following boolean function:

$$F(A, B, C) = \sum m(1, 3, 5, 7).$$

1 = 001	$\bar{A} \cdot \bar{B} \cdot C$
3 = 011	$\bar{A} \cdot B \cdot C$
5 = 101	$A \cdot \bar{B} \cdot C$
7 = 111	$A \cdot B \cdot C$

↑
F=1

"canonical if every cube in the expression has a unique representation in which all of the literals are in complemented or uncomplemented form"

$$\text{SOP form (canonical)} = \bar{A}\bar{B}C + \bar{A}BC + A\bar{B}C + ABC$$

↑
 $\bar{A}\bar{B}C$ ← maybe hard to read?

⑤ Express $G = (A'BCD' + A'BCD + A'B'C'D' + A'B'C'D)'$ as a sum of minterms.

$$\bar{A}BC\bar{D} = 0110_B = 6 \quad \bar{A}BCD = 0111_B = 7$$

$$\bar{A}\bar{B}\bar{C}\bar{D} = 0000_B = 0 \quad \bar{A}\bar{B}\bar{C}D = 0001 = 1$$

As all of the values above where the function returns 0, then G returns 1s at minterms 2, 3, 4, 5.

$$G = \sum m(2, 3, 4, 5)$$

⑧ Find the complement of the following expressions:

8a) $a'b + ab' \rightarrow ab + a'b'$

8b) $b + (cd' + e)a'$
 $= b + a'cd' + a'e \rightarrow b' \cdot (a'cd')' \cdot (a'e)'$

~~$(a' + b + e)(b' + e')(a + e)$~~
 ~~$= (a' + b + e) \cdot (b' + e') \cdot (a + e)$~~

8 continued

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8c) find complement of:

$$(a' + b + c)(b' + c')(a + c)$$

$$\rightarrow (a' + b + c)' + (b' + c')' + (a + c)'$$

10d) Using Karnaugh Map, simplify the following boolean function

$$F(a, b, c, d) = \sum m(0, 1, 2, 3, 4, 5, 6, 8, 9, 12, 13, 14)$$

cd \ ab	00	01	11	10
00	1 ⁰	1 ¹	0 ³	1 ²
01	1 ⁴	1 ⁵	0 ⁷	1 ⁶
11	1 ¹²	1 ¹³	0 ¹⁵	1 ¹⁴
10	1 ⁸	1 ⁹	0 ¹¹	0 ¹⁰

Annotations:
- A vertical group of four 1s in the first column (cd=00, 01, 11, 10) is circled and labeled c' .
- A horizontal group of four 1s in the first row (ab=00, 01, 11, 10) is circled and labeled $a'd'$.
- A horizontal group of two 1s in the last row (ab=10, 11) is circled and labeled bd' .

$$F(a, b, c, d) = c' + a'd' + bd'$$

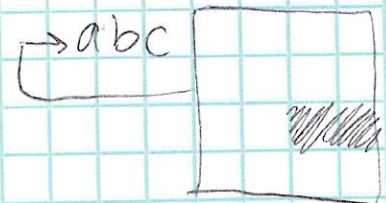
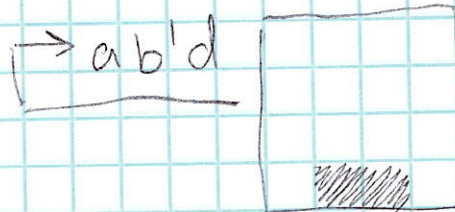
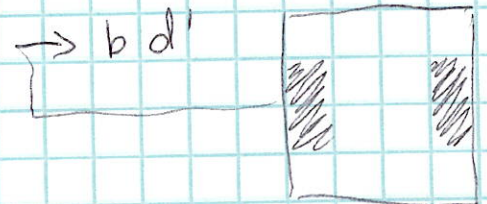
12 12a Draw the K-MAP for $f = \sum m(0, 4, 6, 8, 9, 11, 12, 14, 15)$

cd \ ab	00	01	11	10
00	1 ⁰	0 ¹	0 ³	0 ²
01	1 ⁴	0 ⁵	0 ⁷	1 ⁶
11	1 ¹²	0 ¹³	1 ¹⁵	1 ¹⁴
10	1 ⁸	1 ⁹	1 ¹¹	0 ¹⁰

Annotations for K-map:

- bd' (covers cells 0, 4, 8, 12)
- abc (covers cells 12, 14, 15)
- $c'd'$ (covers cells 0, 1, 2, 3)
- acd (covers cells 12, 13, 14, 15)
- $ab'd$ (covers cells 8, 9, 10, 11)
- $ab'c'$ (covers cells 4, 5, 6, 7)

12b Prime implicants:



12c Identify the essential prime implicants for f .

EPIs cover at least 1 minterm that can't be covered by any other prime implicant.

EPIs : $c'd'$ (covers $m=0$), bd' (covers $m=6$).

12d)

Find all minimal expressions of f and identify those that use only essential prime implicants.

$$f = c'd' + bd' + ab'c' + acd$$

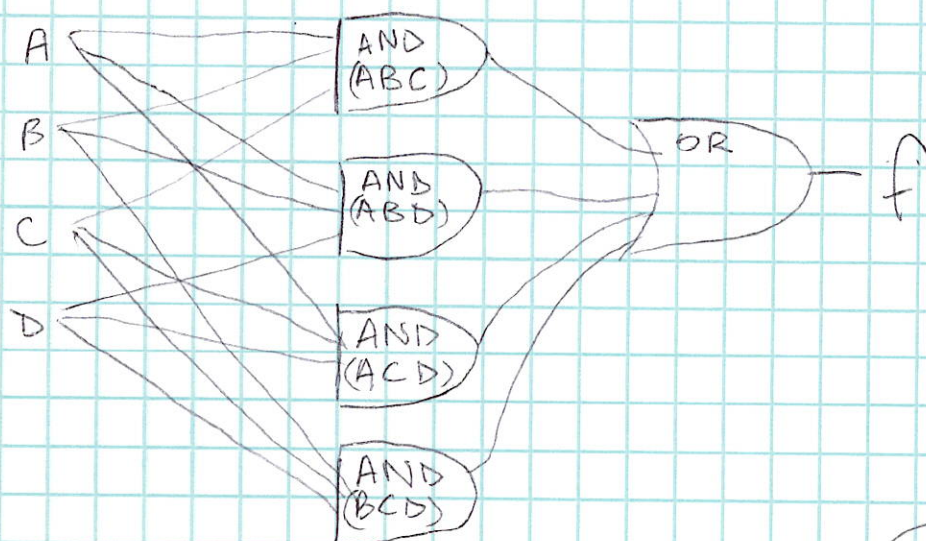
$$f = c'd' + bd' + ab'd + acd$$

$$f = c'd' + bd' + ab'd + abc$$

$c'd'$ and bd' are the only essential prime implicants of f - no representations use ~~only~~ only EPIs.

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Design a 2 level ~~circuit~~ circuit that implements a 4-bit majority function, i.e. returning 1 if 3+ inputs equal 1.

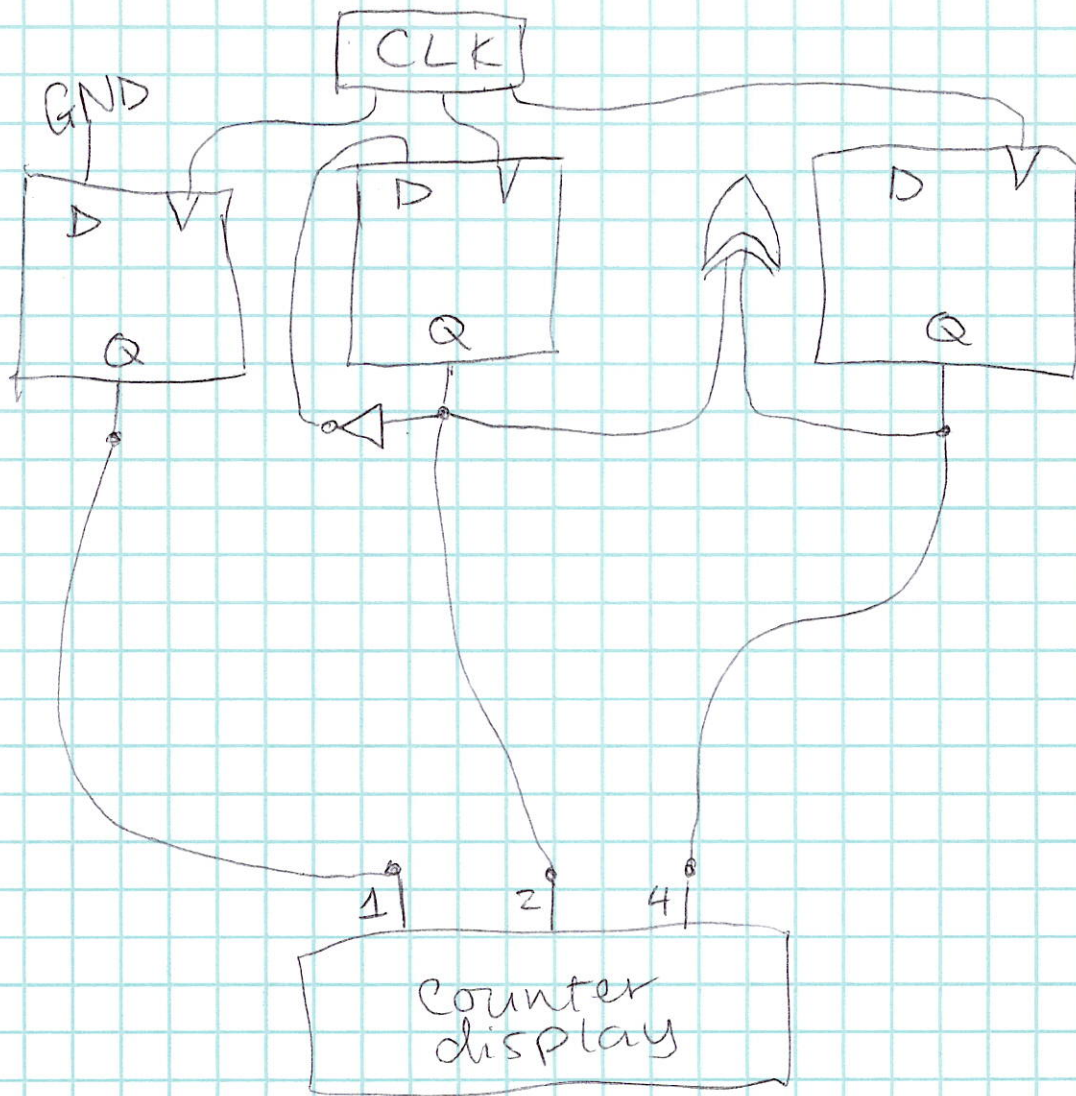


(wires overlap
not touch)

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Q2: Do #3, #4, #6, #7, #14, #15 in Ch 3.

3



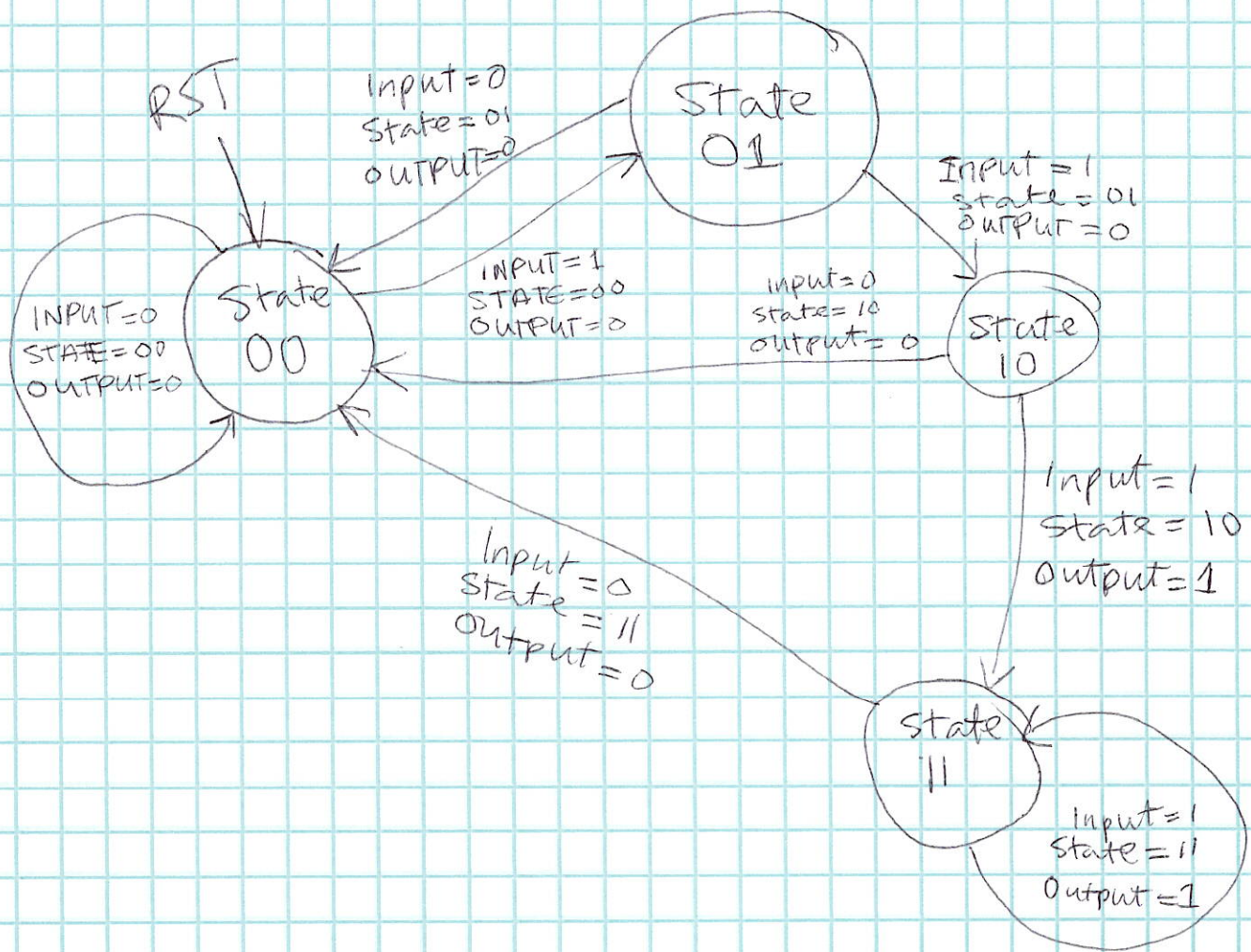
Assume all D type flipflops shown are initialised with the value $Q=0$, maybe through a Reset (Async?)

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- ④ Draw the state transition graph of a mealy machine that samples a serial bit stream and asserts an output if the last 3 samples are 1.

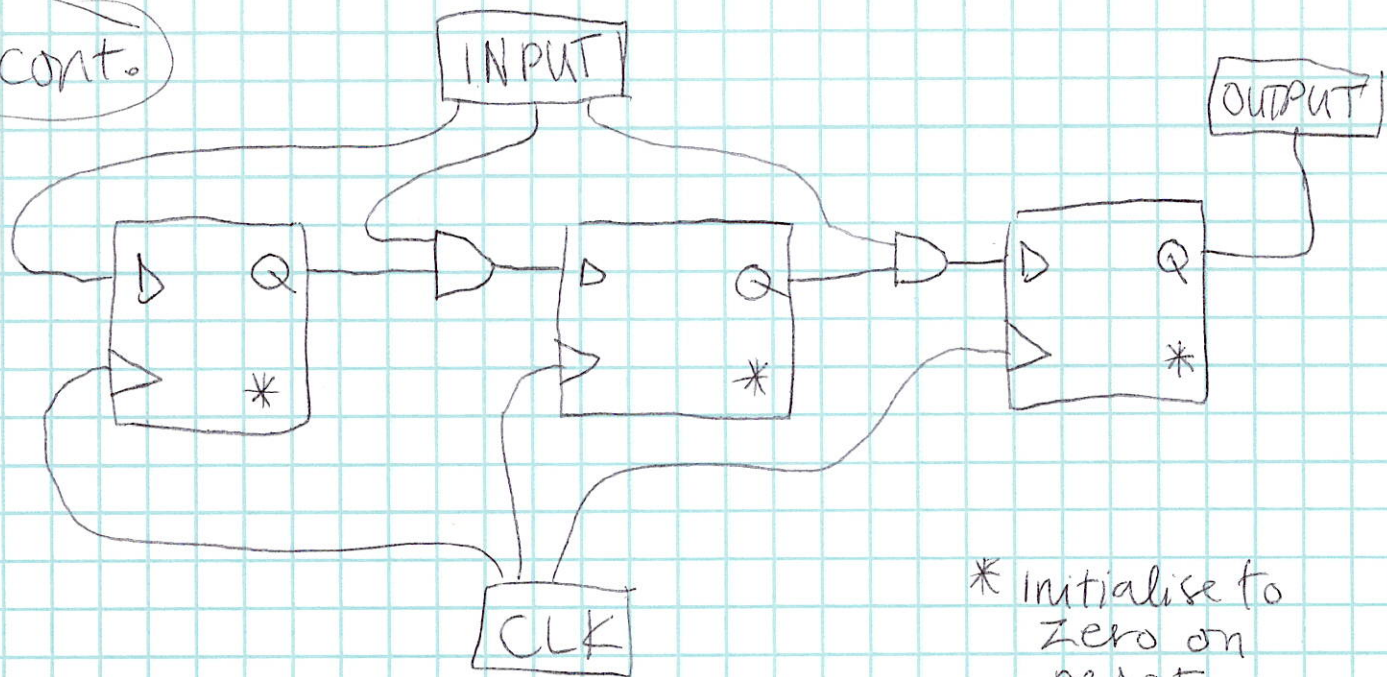
Assume samples made at inactive edges of the clock.

Design the machine with D type flipflops.



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4, cont.



* Initialise to Zero on reset.
(CLR' not shown)

6) Examine Mealy machine described [] & identify possible glitches.

B_{IN}	0	1
$Q_1 Q_0$	0 s=0	1 s=0
01	0 s=1	0 s=1
11	—	—
10	0 s=2	0 s=2

B_{IN}	0	1
$Q_1 Q_0$	1 s=0	0 s=0
01	0 s=1	0 s=1
11	—	—
10	0 s=2	0 s=2

B_1	0	1
$Q_1 Q_0$	0 s=0	1 s=0
01	1 s=1	1 s=1
11	—	—
10	0 s=2	0 s=2

Next $Q_1 = \overline{Q_1} \overline{Q_0} B_{IN}$. Next $Q_0 = \overline{Q_1} \overline{Q_0} B_{IN}$ Bout = $Q_1 Q_0 + \overline{Q_1} B_{IN}$

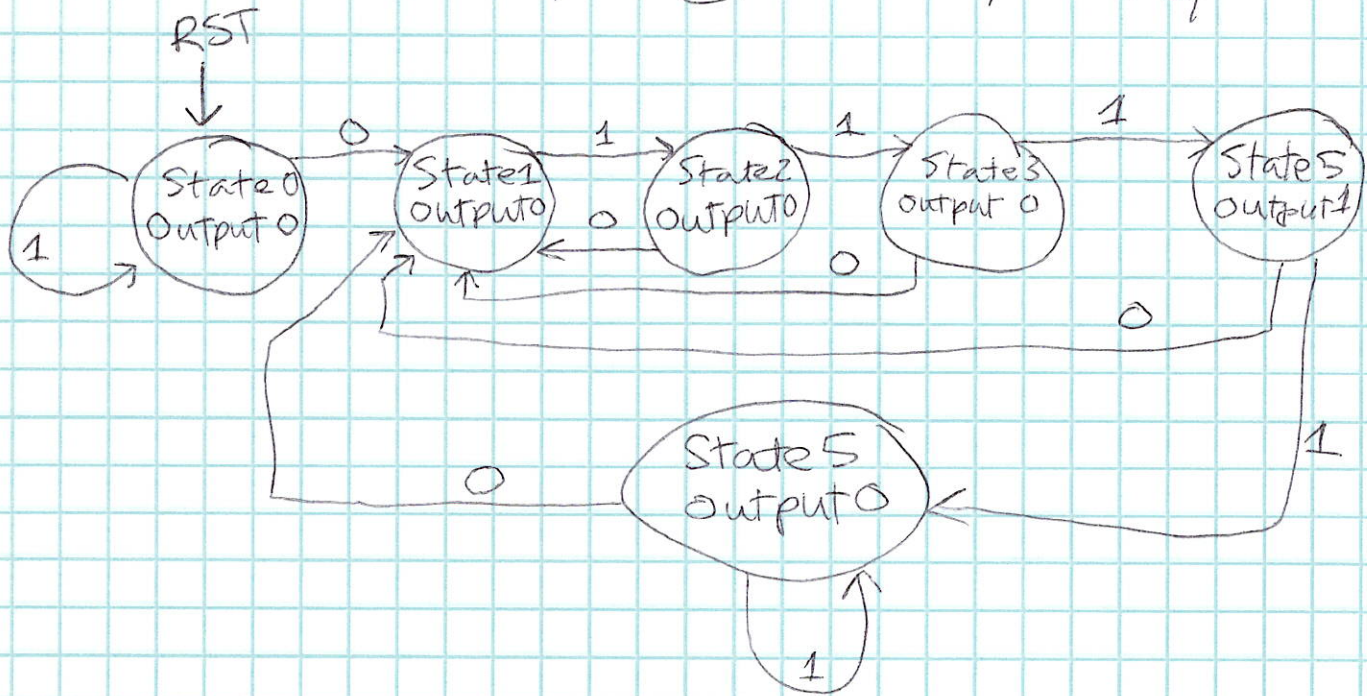
No static 1 glitches (no adjacent non-connected minterm pairs).

~~Don't see static 0 glitches~~

Potential glitch when outputs propagating & when inputs jump from 10 → 01 & vice versa.

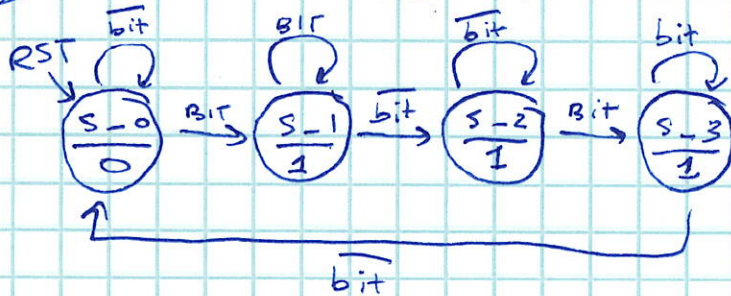
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⑦ Moore FSM, 1 @ 0011, 4 tick, back to 0.



⑭ State Reduction: Equivalent States.

~~Moore FSM~~



(A) S-0 outputs 0
S-2 outputs 1.
not equivalent.

(B) S-1 outputs 1.
S-3 outputs 1.

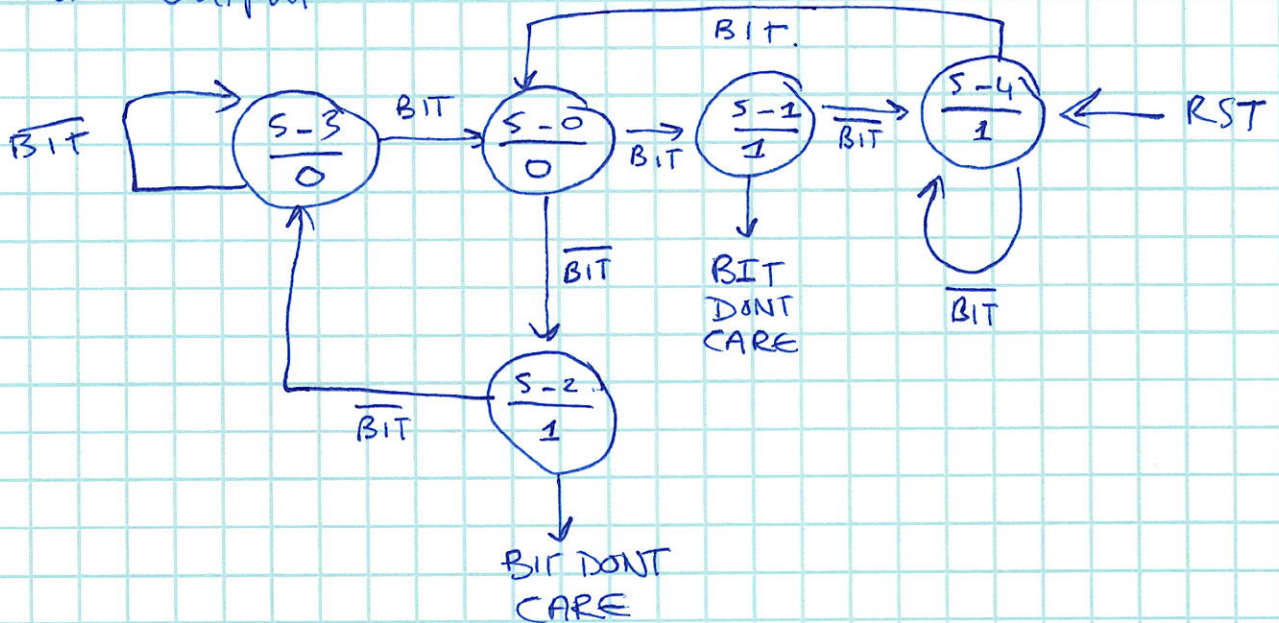
however, when $\overline{\text{bit}} = 1$,
S-1 moves to S-2,
S-3 moves to S-0.
S-2 and S-0 are
not equivalent and
therefore neither are
S-1 and S-3.

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(15) find equivalent states & draw reduced machine.

states where output = 0. $S-0$ outputs 0 and $S-3$ outputs 0. however, when $\text{bit} = 1$, $S-3$ points to $S-0$ and $S-0$ points to $S-1$. $S-0$ and $S-1$ have different outputs so $S-0$ is not equivalent to $S-3$.

all transitions shown for $S-0$ & $S-3$; no 'don't care' outputs.

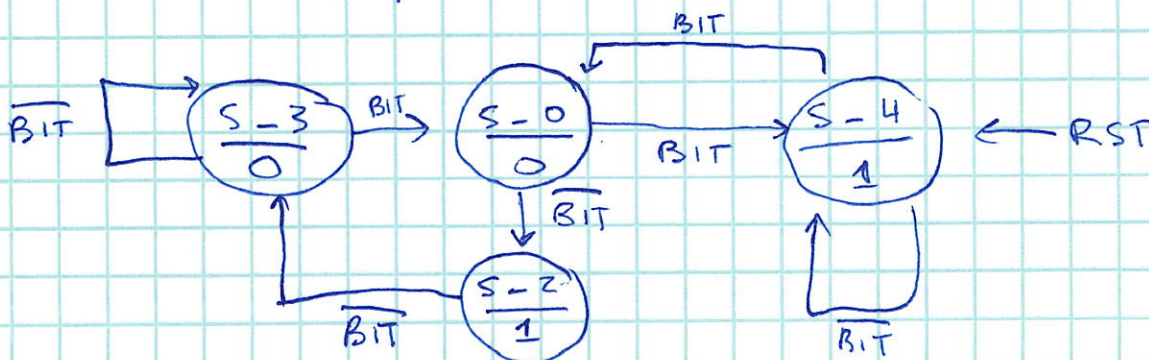


$S-1$ not equivalent to $S-2$: if $\overline{\text{BIT}} = 1$, $S-1 \rightarrow S-4$ (output 1) and $S-2 \rightarrow S-3$ (output 0).

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$S-2$ not equivalent to $S-4$: if $\overline{\text{BIT}} = 1$, $S-2 \rightarrow S-3$ (output 0) and $S-4 \rightarrow S-4$ (output 1).

$S-4$ equivalent to $S-1$. If $\overline{\text{BIT}} = 1$, $S-1 \rightarrow S-4$ (output 1) and $S-4 \rightarrow S-4$ (output 1). When $\text{Bit} = 1$, $S-4 \rightarrow S-0$ (output 0) and $S-1 \rightarrow$ Don't care, which could be made to equal 0.



STG Reduced Machine

Q3: Convert following numbers into decimal.
assume unsigned numbers.

(a) $1010100_2 = 4 + 16 + 64 = 84_{\text{base 10}}$

$\uparrow \quad \uparrow \quad \uparrow$
 $64 \quad 16 \quad 4$

(b) $1111000_2 = 64 + 32 + 16 + 8 = 120_{\text{base 10}}$

$\uparrow \uparrow \uparrow \uparrow$
 $64 \quad 32 \quad 16 \quad 8$

$\overbrace{80}$
 $\underbrace{40}$

(c) $3221_4 = 1 + 8 + 32 + 192 = 233_{\text{base 10}}$

$\uparrow \quad \uparrow \quad \uparrow \quad \uparrow$
 $1 \quad 8 \quad 2 \times 4 \quad 1$
 $2 \times 16 = 32$
 $3 \times 64 = 192$

$\overbrace{40}$
 $\underbrace{193}$

$\begin{array}{r} 64 \\ 3 \\ \hline 192 \end{array}$

(d) $AA5F_{16} = 15 + 80 + 2560 + 40960 = 43615_{\text{base 10}}$

$\uparrow \quad \uparrow \quad \uparrow \quad \uparrow$
 $15 \quad 5 \times 16 = 80$
 $10 \times 16^2 = 10 \times 256$
 $10 \times 16^3 = 4096 \times 10$

$\begin{array}{r} 40960 \\ 2560 \\ 80 \\ + 15 \\ \hline 43615 \end{array}$

Q4: Show following numbers in 2's complement binary then add them.

(a) $-47_{10} = -1(00101111)$

$\begin{array}{ccccccc} & 32 & 16 & 8 & 4 & 2 & 1 \\ 0 & 0 & 1 & 0 & 1 & 1 & 1 & 1 \end{array}$

1's complement: 11010000.

2's complement: 11010001 (add 1)

(b) $-123_4 = (16 + 8 + 3) \times -1 = -27_{\text{dec.}}$

$27_{10} = 00011011$

keep last bit, flip rest:

$-123_4 = 11100101$

see next page for addition.

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Q 4, cont.

$$-47_{10} = 110100101$$

$$-123_4 = 11100101 +$$

$$1 \quad 101 \quad 1011$$

↑
carry bit,
ignore.

$$\text{sum} = 10110110$$

~~$$\text{sum} = 10110110$$~~

Q 5: Use tabular method to minimize expression.

A. Identify prime implicants.

B. Create minterm table assignments to prime implicants.

C. Identify & select essential prime implicants

D. Select remaining prime implicants & create minimal minterm cover.

assuming full stops
intended as commas?

$$Z(a,b,c,d) = a'b'(c'd + c'd') + a'bc'd + ab(c+d) + ac'd$$

$$= a'b'c'd + a'b'c'd' + a'bc'd + abc + abd + ac'd$$

→ $a'b'c'$
d & d' mentioned.

~~$a'b'c'd$ subset of $a'b'c'$~~

WRONG! see next page sorry

~~$$abc + abd + ac'd$$~~

ab \ cd	00	01	11	10
00	0	0	1	0
01	0	1	1	0
11	1	1	1	1
10	1	0	1	0

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$$\begin{aligned}
 Q5: \quad z &= a'b'(c'd + c'd') + a'bc'd + ab(ctd) + ac'd. \\
 &= \cancel{a'b'c'} + a'bc'd + \cancel{abc} + \cancel{abd} + ac'd. \\
 &= a'b'c'd + a'b'c'd' + a'bc'd + ac'd + abc + abd.
 \end{aligned}$$

$$= c'd(a'b' + a'b + a) + a'b'c'd' + abc + abd.$$

$$= c'd + a'bc'd' + abc + abd + abc'd.$$

$$= c'd + a'b'c'd' + abc + abd + abc'd.$$

$$= c'd + a'b'c'd' + abc + abc'd'$$

$$= c'd + a'b'c' + abc \leftarrow \text{Prime Implicants}$$

DOUBLE CHECK

	$c'd$	$a'b'c'$	abc
$a'b'c'd$	X	X	
$a'b'c'd'$		X	
$a'bc'd$	X		
$abc'd'$			X
$abcd$			X
$abc'd$	X		
$abc'd$	X		
$ab'c'd$	X		

← alone in row

← alone

← alone

All are essential prime implicants?

No Remaining prime implicants?

minimal minterm cover =
 $c'd + a'b'c' + abc$.

$x=1$

	00	01	11	10
00	X	X		
01		X		
11		X	X	X
10		X		

add $[abd]$
to avoid
glitching

Q6: 3 to 4 sentences of current proj. ideas.

I plan to work with Francis & Tanner, and we will try to find a fourth after meeting on sundays to discuss project ideas.

After we met, we ~~will do~~ ^{decided on} a project; To build a FIR (Finite Impulse Response) filter in Verilog.

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